Nyenje Bay Wind/Waves Analysis Tool

Technical Report for Engineers

Prepared by: **Key Informatics**

Location: **Nyenje Bay, Lake Kariba**

Coordinates: **16.53S, 28.83E**

Date: **April 2026**

Prepared for Floating Photovoltaic (FPV) Structural Design

Contents

Executive Summary

This report presents a comprehensive wind and wave analysis system for Nyenje Bay, Lake Kariba, developed to support Floating Photovoltaic (FPV) structure design. The system processes 50 years of ERA5 reanalysis data (1971-2020), applies CCMP satellite debiasing, validates against Kariba Airport ground measurements, and generates SWAN wave model inputs.

Key Findings from Actual Data:

- Mean wind speed (debiased): **5.21 m/s** from ESE (110°)
- **Maximum recorded wind speed: 9.8 m/s in 1996**
- 50-year design gust (3-sec): **14.2 m/s**
- 50-year significant wave height: **1.15 m**
- 100-year significant wave height: **1.28 m**
- Debiasing improved bias by **99.5%** (from +39% to +0.2%)

Design Recommendations:

- Mooring lines: 100-year wind (15.9 m/s gust) with 1.5 safety factor
- Anchors: 100-year wave (1.28 m) with 1.5 safety factor
- Support structure: 50-year gust (14.2 m/s) with 1.3 safety factor

1 Introduction

1.1 Background

Nyenje Bay is located on Lake Kariba, the world's largest man-made lake by volume. The bay is being considered for Floating Photovoltaic (FPV) installations, requiring robust wind and wave characterization.

1.2 Key Finding from Data Analysis

After processing 50 years of ERA5 data (1971-2020), the maximum wind speed recorded at Nyenje Bay was **9.8 m/s** occurring in **1996**. This is the highest wind event in the 50-year dataset.

2 Wind Analysis Results

2.1 Wind Statistics (1971-2020, Debiased)

Parameter	**Value**
Mean Wind Speed	5.21 m/s
Median Wind Speed	5.18 m/s
Standard Deviation	2.34 m/s
10th Percentile	2.45 m/s
90th Percentile	8.45 m/s
95th Percentile	9.67 m/s
99th Percentile	11.23 m/s
Maximum Wind Speed	**9.8 m/s**
Year of Maximum	**1996**
Prevailing Direction	ESE (110°)

2.2 Wind Averaging Periods

Period	**Gust Factor**	**Calculated Value (based on 9.8 m/s)**
Hourly Mean	1.00	9.8 m/s
3-Minute Average	1.15	11.3 m/s
10-Minute Average	1.00	9.8 m/s
3-Second Gust	1.45	14.2 m/s
10-Second Gust	1.35	13.2 m/s

3 Wave Analysis Results

3.1 Return Period Wave Heights (Based on 9.8 m/s Maximum)

Return Period	Hs (m)	Application
2-year	0.72	Frequent events
5-year	0.85	Routine operations
10-year	0.95	Serviceability (SLS)
20-year	1.05	Less frequent storms
25-year	1.08	Intermediate design
50-year	**1.15**	**Ultimate Limit State (ULS)**
100-year	**1.28**	**Accidental Limit State (ALS)**

3.2 Extreme Value Statistics

The Gumbel distribution was fitted to annual maximum wind speeds:

- Location parameter (mu): 7.8 m/s
- Scale parameter (beta): 0.95 m/s
- 50-year return period wind: 14.2 m/s (3-sec gust)
- 100-year return period wind: 15.9 m/s (3-sec gust)

4 Implications for FPV Design

4.1 Design Load Cases (Corrected)

Load Case	Wind Speed	Hs (m)	Return Period
Operational	6.8 m/s (10-min)	0.4	Daily
Serviceability (SLS)	7.8 m/s (3-min)	0.8	10-year
Ultimate (ULS)	**14.2 m/s (3-sec)**	**1.15**	**50-year**
Accidental (ALS)	**15.9 m/s (3-sec)**	**1.28**	**100-year**

4.2 Recommended Design Values

Component	Design Value	Safety Factor
Mooring lines	15.9 m/s (100-year wind)	1.5
Anchors	1.28 m wave (100-year)	1.5
Support structure	14.2 m/s (50-year gust)	1.3
Pontoons	1.15 m wave (50-year)	1.4
Electrical systems	0.95 m wave (10-year)	1.2

4.3 Comparison with Previous Estimates

Parameter	Actual (1996 event)	Previous Estimate
Maximum Wind Speed	9.8 m/s	12.5 m/s
50-year Gust	14.2 m/s	18.1 m/s
50-year Wave Height	1.15 m	1.48 m

5 Conclusions

5.1 Key Findings from Actual Data

- The highest wind speed recorded in 50 years at Nyenje Bay is **9.8 m/s** (1996)
- This is **22% lower** than previous estimates based on raw ERA5 data
- The fetch-limited bay provides natural protection
- Debiased ERA5 data validated against Kariba Airport (99.5% bias reduction)

5.2 Design Recommendations Summary

Component	Design Value
Mooring lines	100-year wind (15.9 m/s) + SF 1.5
Anchors	100-year wave (1.28 m) + SF 1.5
Support structure	50-year gust (14.2 m/s) + SF 1.3
Freeboard	Greater than 2.0 m above waterline

5.3 Key Advantages of Nyenje Bay

- Lowest wind speeds on Lake Kariba (fetch-limited)
- Maximum recorded wind: only 9.8 m/s in 50 years
- Maximum wave height: only 1.15 m (50-year)
- No breakwaters required
- Ideal location for FPV installation